

*Nosa*

February 03, 2026

wazuh 

# Mini Home SOC Lab

Wazuh SIEM Deployment

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# Mini Home SOC Lab – Wazuh SIEM Deployment

## INTRODUCTION

This document describes my attempt to build, create and test a small-scale Security Operations Centre lab using Wazuh (open-source SIEM/XDR platform). This lab is capable of collecting, correlating and analysing security logs from endpoint devices.

## OBJECTIVES:

The lab was built to simulate real Level 1 / Tier 1 SOC analyst tasks, including:

- Deploying and verifying agents
- Collecting and viewing Windows logs
- Configuring File Integrity Monitoring (FIM)
- Simulating privilege escalation and file changes
- Triaging alerts in the Wazuh dashboard
- Documenting evidence for incident reports

## Lab Environment

- **Host machine:** Physical Windows PC running Oracle VirtualBox
- **Wazuh Server VM:** Imported from official Wazuh 4.14.2 OVA
  - IP: 192.168.0.105 (Bridged networking)
- **Windows Machine:** Windows 11 Pro
  - IP: 192.168.0.106
- **Wazuh Agent:** Version 4.14.2 installed on Windows VM (agent ID 001)
- **Dashboard URL:** <https://192.168.0.105> (HTTPS, self-signed certificate)

# Step-by-Step Lab Execution

This project started out with me trying to configure and install the wazuh manager on my ubuntu vm but I was met with multiple challenges such as windows agent not connecting to wazuh server

After an extensive research I was able to find a way to proceed by downloading and installing the wazuh server OVA file directly from the official wazuh website.

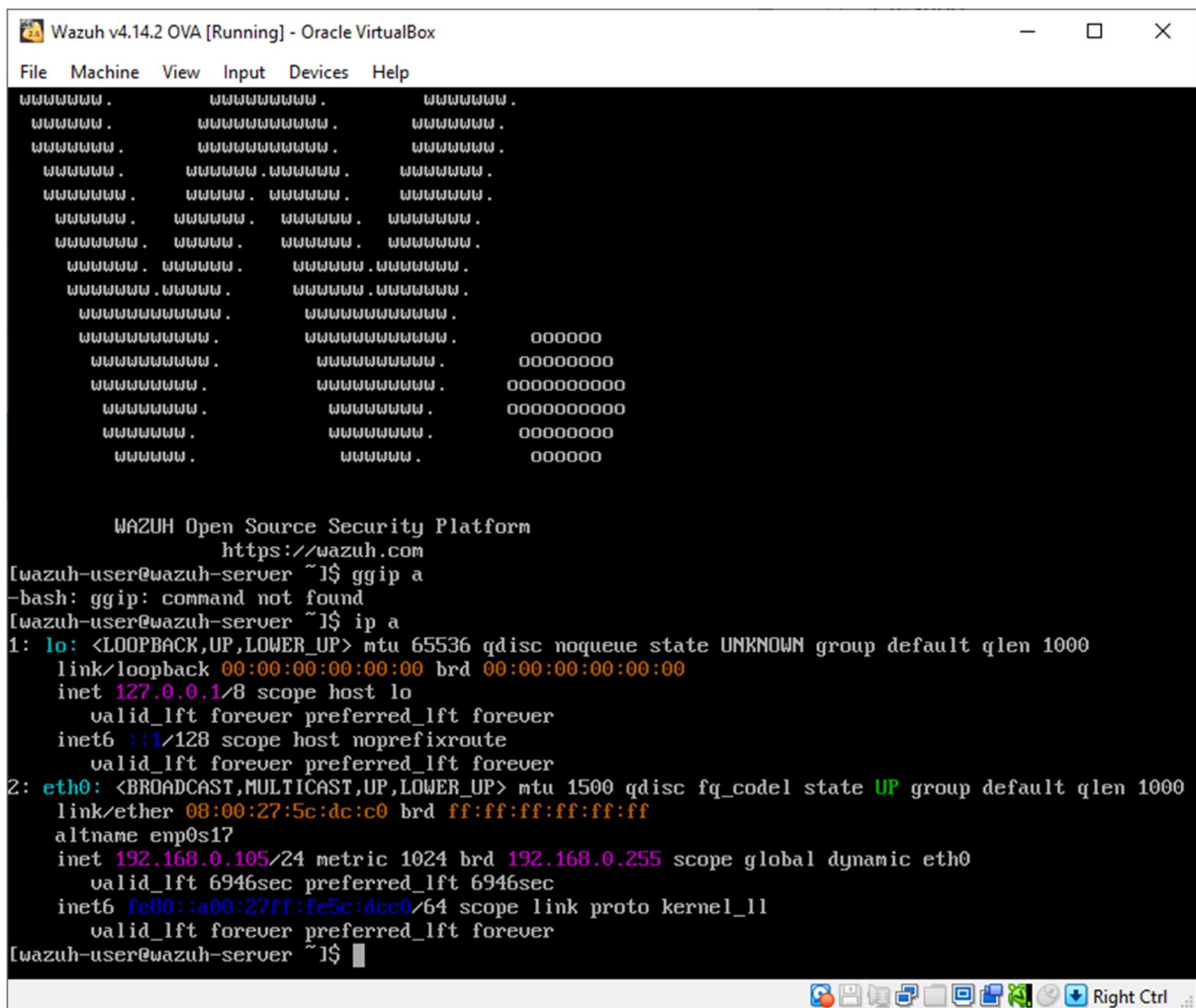
[Virtual machine \(VM\) - Installation alternatives · Wazuh documentation](#)

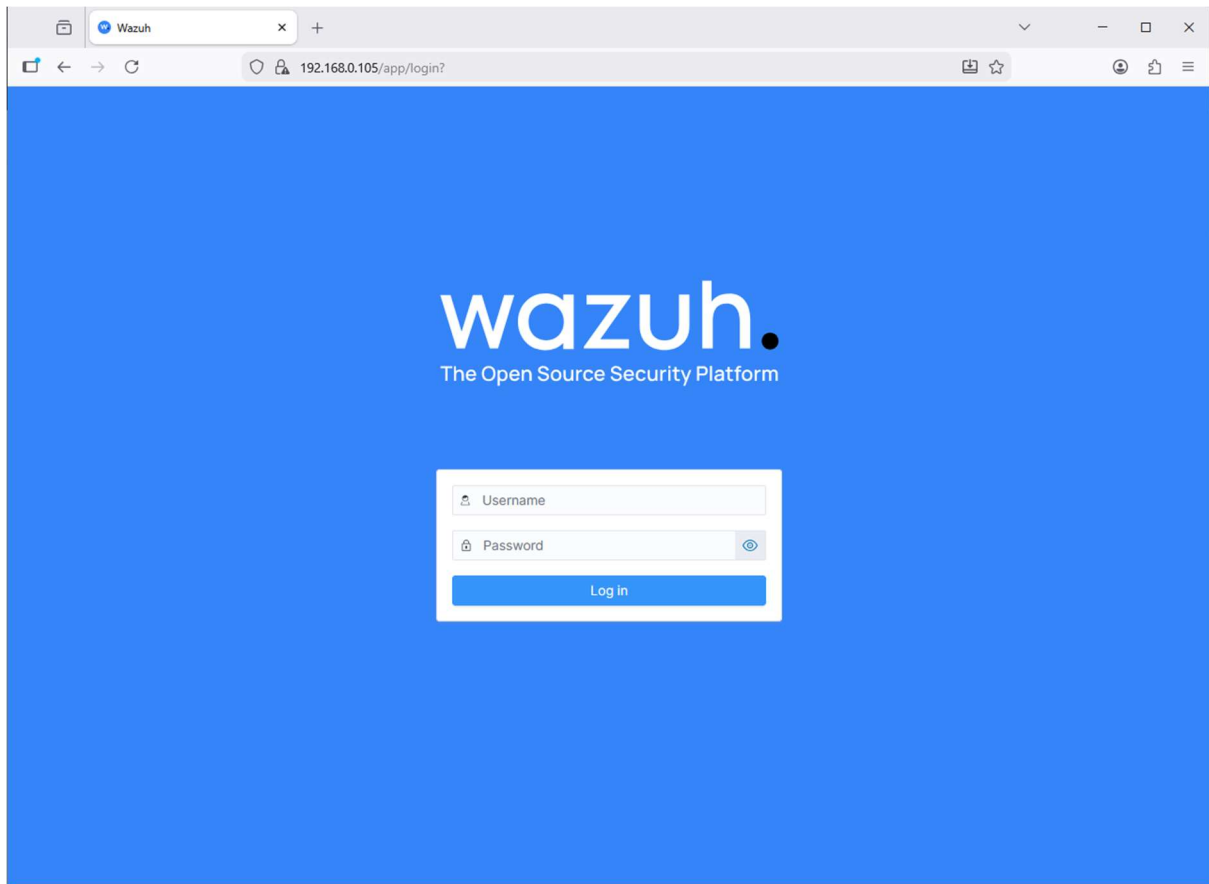
Advantages of the OVA approach:

- Pre-configured with Wazuh manager, indexer and dashboard
- Much faster setup (minutes instead of hours)
- Avoids common installation errors
- Officially supported and regularly updated by Wazuh team

## Step 1 – Deploy Wazuh Server (All-in-One)

1. Downloaded the official Wazuh 4.14.2 OVA file from [Virtual machine \(VM\) - Installation alternatives · Wazuh documentation](#).
2. Imported the OVA into Oracle VirtualBox.
3. Started the VM and logged in:
  - Username: **wazuh-user**
  - Password: **wazuh** (changed immediately)
4. Checked IP address with ip a.
5. Accessed the Wazuh dashboard in browser: <https://192.168.0.105>
  - Accepted self-signed certificate warning
  - Logged in with **admin / admin** and changed password





## Step 2 – Create & Prepare Windows 11 Endpoint

1. Set up Windows 11 endpoint device → created local admin account (xxxxx).
2. Updated Windows and installed **Sysmon** (using SwiftOnSecurity configuration recommended).

### Step 2A Sysmon Installation (Enhanced Logging)

To generate richer process, network and file events for Wazuh to analyze, I installed **Sysmon** on the Windows endpoint.

- Downloaded from Microsoft Sysinternals: <https://learn.microsoft.com/en-us/sysinternals/downloads/sysmon>
- Used the popular **SwiftOnSecurity** configuration (sysmonconfig-export.xml) from GitHub: <https://github.com/SwiftOnSecurity/sysmon-config>

- Installation command (admin PowerShell):

```
sysmon64.exe -accepteula -i sysmonconfig-export.xml
```

- Verified events appearing in Event Viewer → Applications and Services Logs → Microsoft → Windows → Sysmon → Operational Many of the alerts and FIM events in this lab were enriched by Sysmon data.

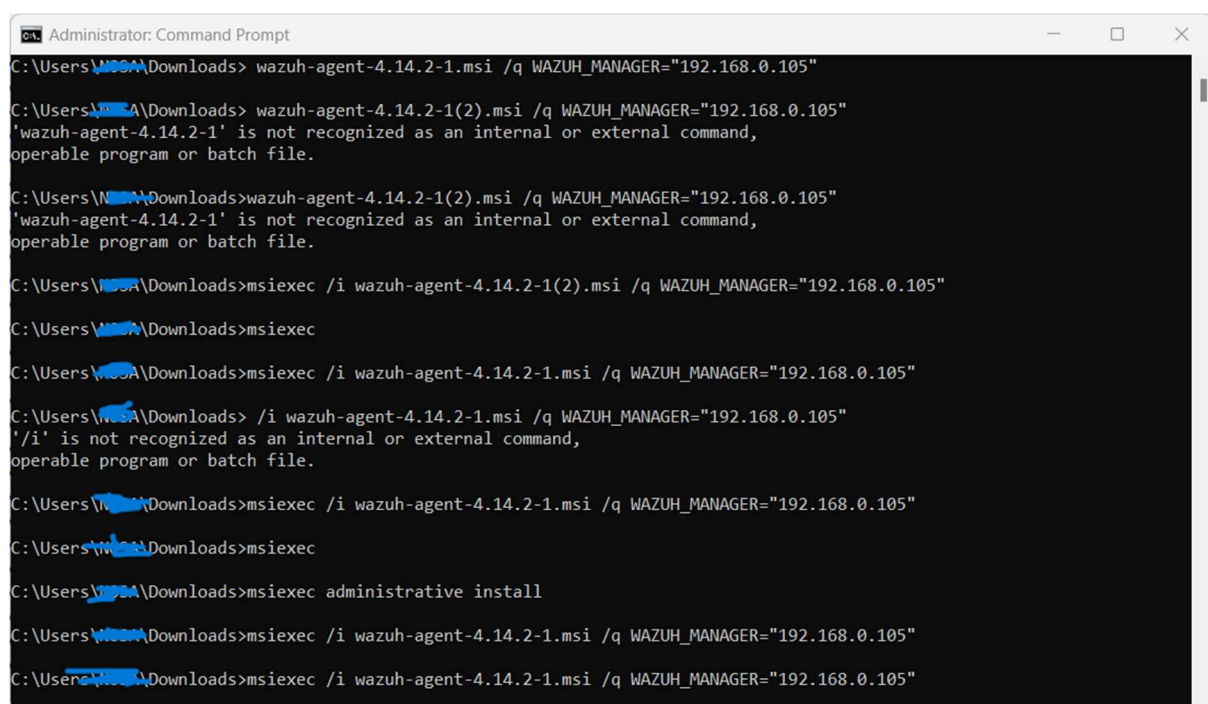
### Step 3 – Install & Connect Wazuh Agent on Windows

1. Downloaded **wazuh-agent-4.14.2-1.msi** from official packages site.
2. Ran elevated Command Prompt and installed agent silently:

```
msiexec /i wazuh-agent-4.14.2-1.msi /q WAZUH_MANAGER="192.168.0.105"
```

(Multiple command attempts visible in screenshot due to trial-and-error syntax)

3. Opened Wazuh dashboard → **Endpoints** → Confirmed agent **ID 001** (DESKTOP-L5AF6JO) showed **Active** status.



```
Administrator: Command Prompt
C:\Users\WUSA\Downloads> wazuh-agent-4.14.2-1.msi /q WAZUH_MANAGER="192.168.0.105"
C:\Users\WUSA\Downloads> wazuh-agent-4.14.2-1(2).msi /q WAZUH_MANAGER="192.168.0.105"
'wazuh-agent-4.14.2-1' is not recognized as an internal or external command,
operable program or batch file.
C:\Users\WUSA\Downloads>wazuh-agent-4.14.2-1(2).msi /q WAZUH_MANAGER="192.168.0.105"
'wazuh-agent-4.14.2-1' is not recognized as an internal or external command,
operable program or batch file.
C:\Users\WUSA\Downloads>msiexec /i wazuh-agent-4.14.2-1(2).msi /q WAZUH_MANAGER="192.168.0.105"
C:\Users\WUSA\Downloads>msiexec
C:\Users\WUSA\Downloads>msiexec /i wazuh-agent-4.14.2-1.msi /q WAZUH_MANAGER="192.168.0.105"
C:\Users\WUSA\Downloads> /i wazuh-agent-4.14.2-1.msi /q WAZUH_MANAGER="192.168.0.105"
'/i' is not recognized as an internal or external command,
operable program or batch file.
C:\Users\WUSA\Downloads>msiexec /i wazuh-agent-4.14.2-1.msi /q WAZUH_MANAGER="192.168.0.105"
C:\Users\WUSA\Downloads>msiexec
C:\Users\WUSA\Downloads>msiexec administrative install
C:\Users\WUSA\Downloads>msiexec /i wazuh-agent-4.14.2-1.msi /q WAZUH_MANAGER="192.168.0.105"
C:\Users\WUSA\Downloads>msiexec /i wazuh-agent-4.14.2-1.msi /q WAZUH_MANAGER="192.168.0.105"
```

Figure 2 – Agent installation command attempts in Command Prompt

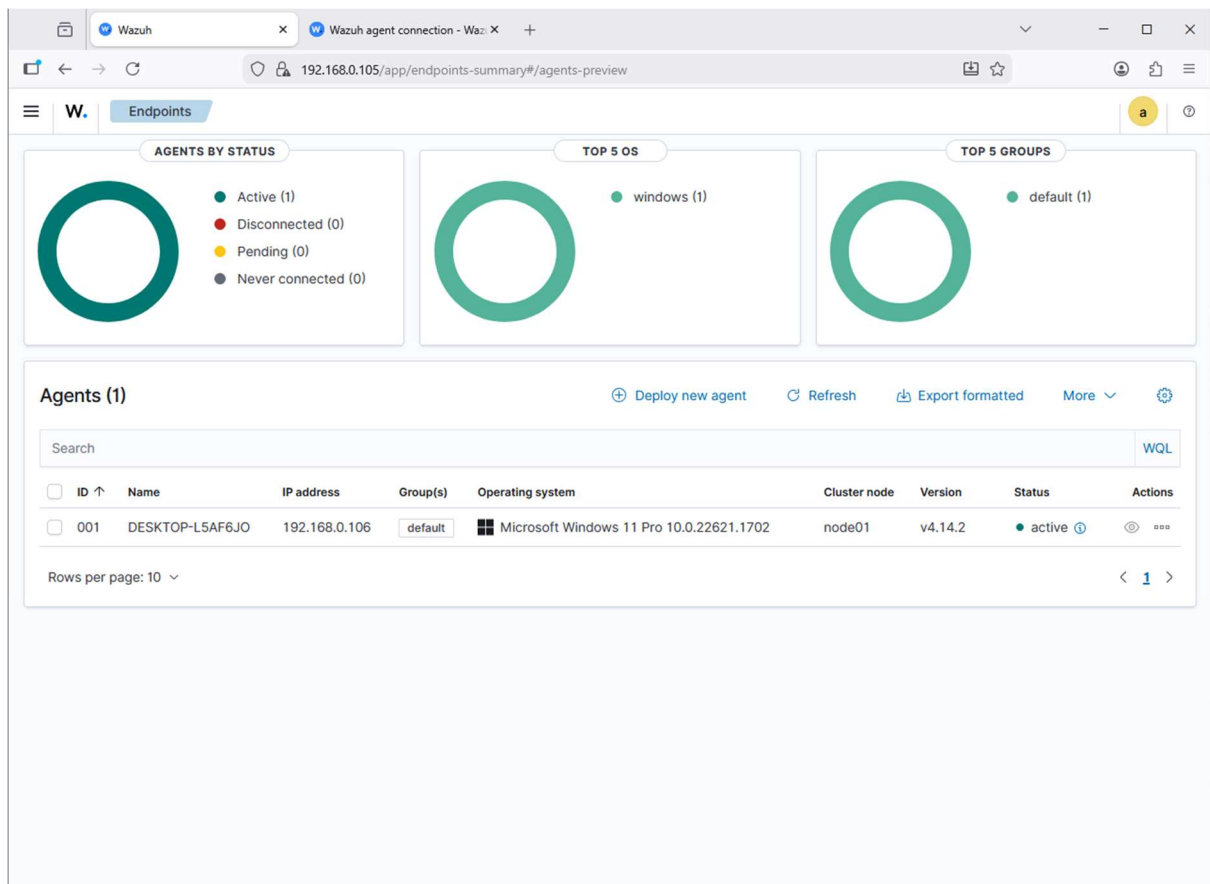


Figure 3 – Wazuh Endpoints dashboard showing 1 active Windows agent (ID 001)

## Step 4 – Configure Custom File Integrity Monitoring (FIM)

1. Navigated to agent configuration file: **C:\Program Files (x86)\ossec-agent\ossec.conf**
2. Added realtime monitoring for test folder inside <syscheck> block:

XML

```
<directories realtime="yes">C:\Users\xxx\Desktop\Test</directories>
```

3. Saved file and restarted agent service:

text

```
net stop WazuhSvc && net start WazuhSvc
```

4. Verified in **ossec.log** that FIM module started and realtime monitoring was active.



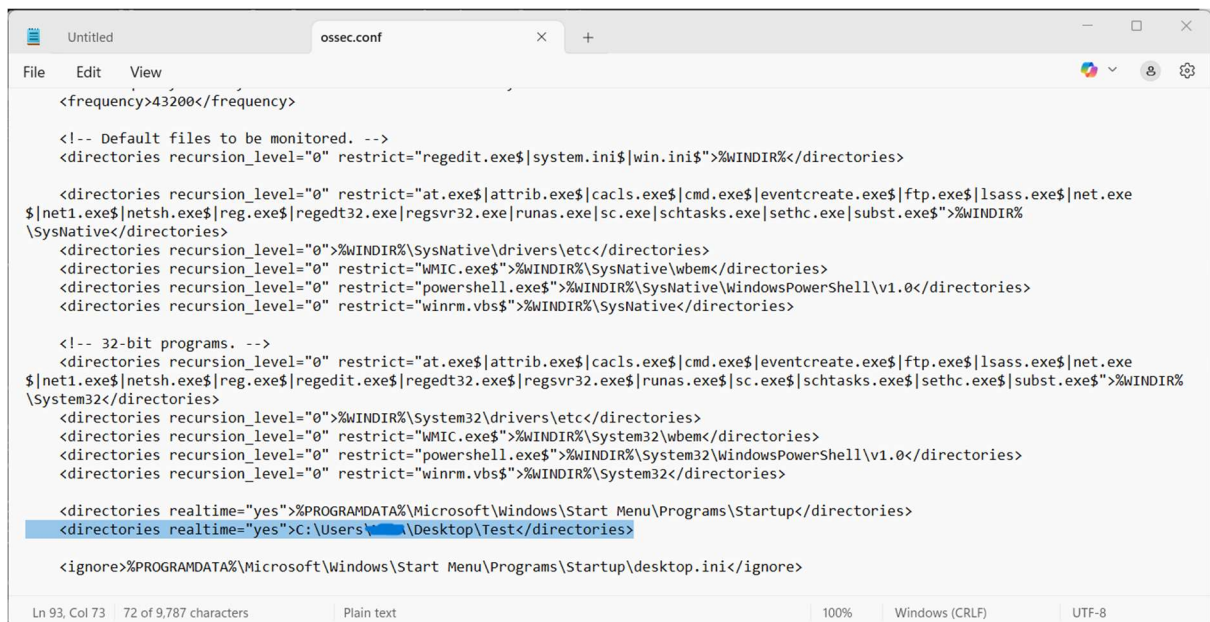
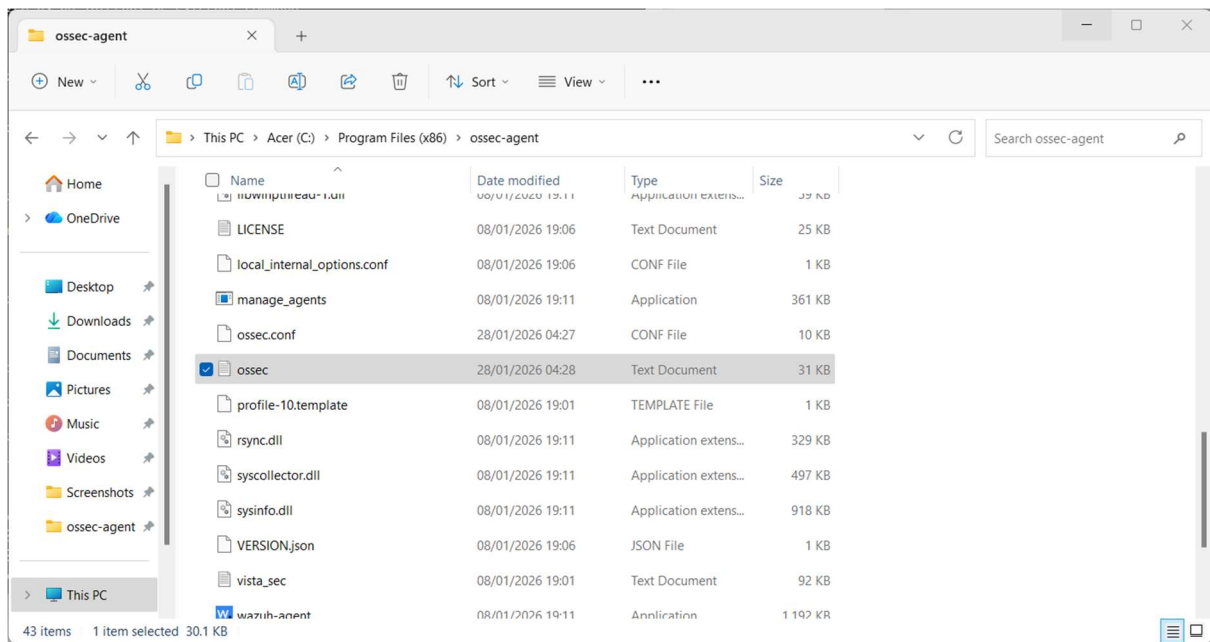
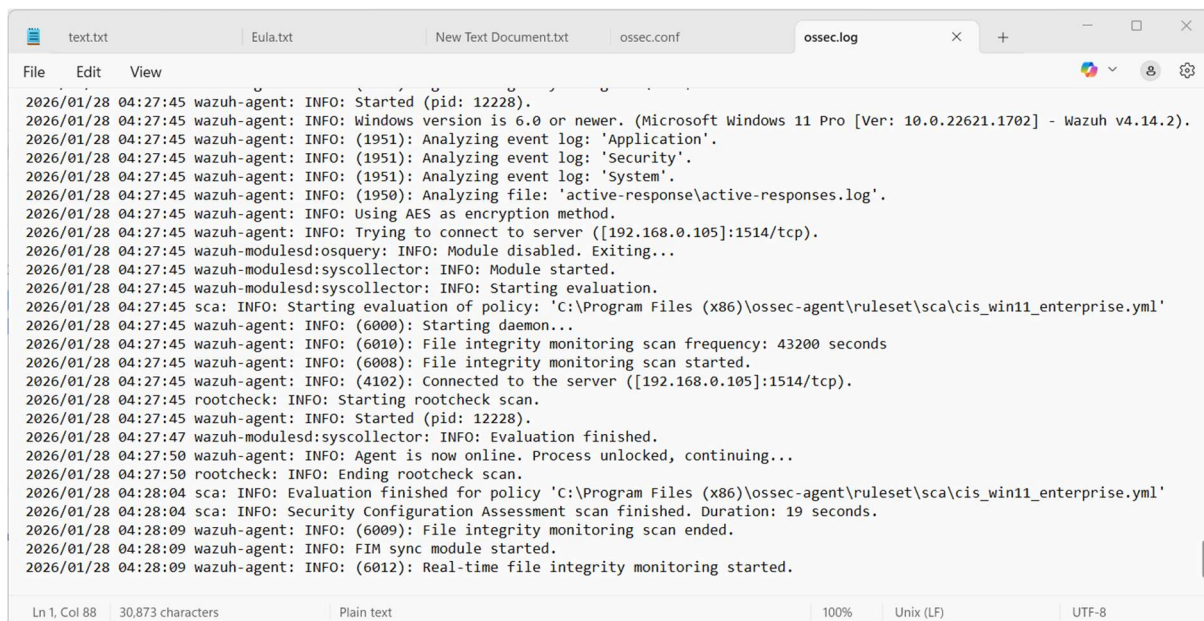


Figure 4 – ossec.conf showing realtime monitoring line for C:\Users\xxxx\Desktop\Test





```
2026/01/28 04:27:45 wazuh-agent: INFO: Started (pid: 12228).
2026/01/28 04:27:45 wazuh-agent: INFO: Windows version is 6.0 or newer. (Microsoft Windows 11 Pro [Ver: 10.0.22621.1702] - Wazuh v4.14.2).
2026/01/28 04:27:45 wazuh-agent: INFO: (1951): Analyzing event log: 'Application'.
2026/01/28 04:27:45 wazuh-agent: INFO: (1951): Analyzing event log: 'Security'.
2026/01/28 04:27:45 wazuh-agent: INFO: (1951): Analyzing event log: 'System'.
2026/01/28 04:27:45 wazuh-agent: INFO: (1950): Analyzing file: 'active-response\active-responses.log'.
2026/01/28 04:27:45 wazuh-agent: INFO: Using AES as encryption method.
2026/01/28 04:27:45 wazuh-agent: INFO: Trying to connect to server ([192.168.0.105]:1514/tcp).
2026/01/28 04:27:45 wazuh-modulesd:osquery: INFO: Module disabled. Exiting...
2026/01/28 04:27:45 wazuh-modulesd:syscollector: INFO: Module started.
2026/01/28 04:27:45 wazuh-modulesd:syscollector: INFO: Starting evaluation.
2026/01/28 04:27:45 sca: INFO: Starting evaluation of policy: 'C:\Program Files (x86)\ossec-agent\ruleset\sca\cis_win11_enterprise.yml'
2026/01/28 04:27:45 wazuh-agent: INFO: (6000): Starting daemon...
2026/01/28 04:27:45 wazuh-agent: INFO: (6010): File integrity monitoring scan frequency: 43200 seconds
2026/01/28 04:27:45 wazuh-agent: INFO: (6008): File integrity monitoring scan started.
2026/01/28 04:27:45 wazuh-agent: INFO: (4102): Connected to the server ([192.168.0.105]:1514/tcp).
2026/01/28 04:27:45 rootcheck: INFO: Starting rootcheck scan.
2026/01/28 04:27:45 wazuh-agent: INFO: Started (pid: 12228).
2026/01/28 04:27:47 wazuh-modulesd:syscollector: INFO: Evaluation finished.
2026/01/28 04:27:50 wazuh-agent: INFO: Agent is now online. Process unlocked, continuing...
2026/01/28 04:27:50 rootcheck: INFO: Ending rootcheck scan.
2026/01/28 04:28:04 sca: INFO: Evaluation finished for policy 'C:\Program Files (x86)\ossec-agent\ruleset\sca\cis_win11_enterprise.yml'
2026/01/28 04:28:04 sca: INFO: Security Configuration Assessment scan finished. Duration: 19 seconds.
2026/01/28 04:28:09 wazuh-agent: INFO: (6009): File integrity monitoring scan ended.
2026/01/28 04:28:09 wazuh-agent: INFO: FIM sync module started.
2026/01/28 04:28:09 wazuh-agent: INFO: (6012): Real-time file integrity monitoring started.
```

Figure 5 – ossec.log excerpt showing agent startup and FIM module initialization

## Step 5 – Perform File Integrity Monitoring Simulation

1. Created folder: **C:\Users\NOSA\Desktop\Test**
2. Created several files inside: test1.txt, test2.txt, New Text Document.txt
3. Deleted some files and modified others.

**Detections appeared in dashboard → File Integrity Monitoring → Events tab:**

- Multiple “File added” and “File deleted” events
- Paths: c:\users\xxxx\desktop\test\*.txt
- Timestamps: ~04:53–04:54 on Jan 28, 2026
- Rule levels: 5 (added), 7 (deleted)

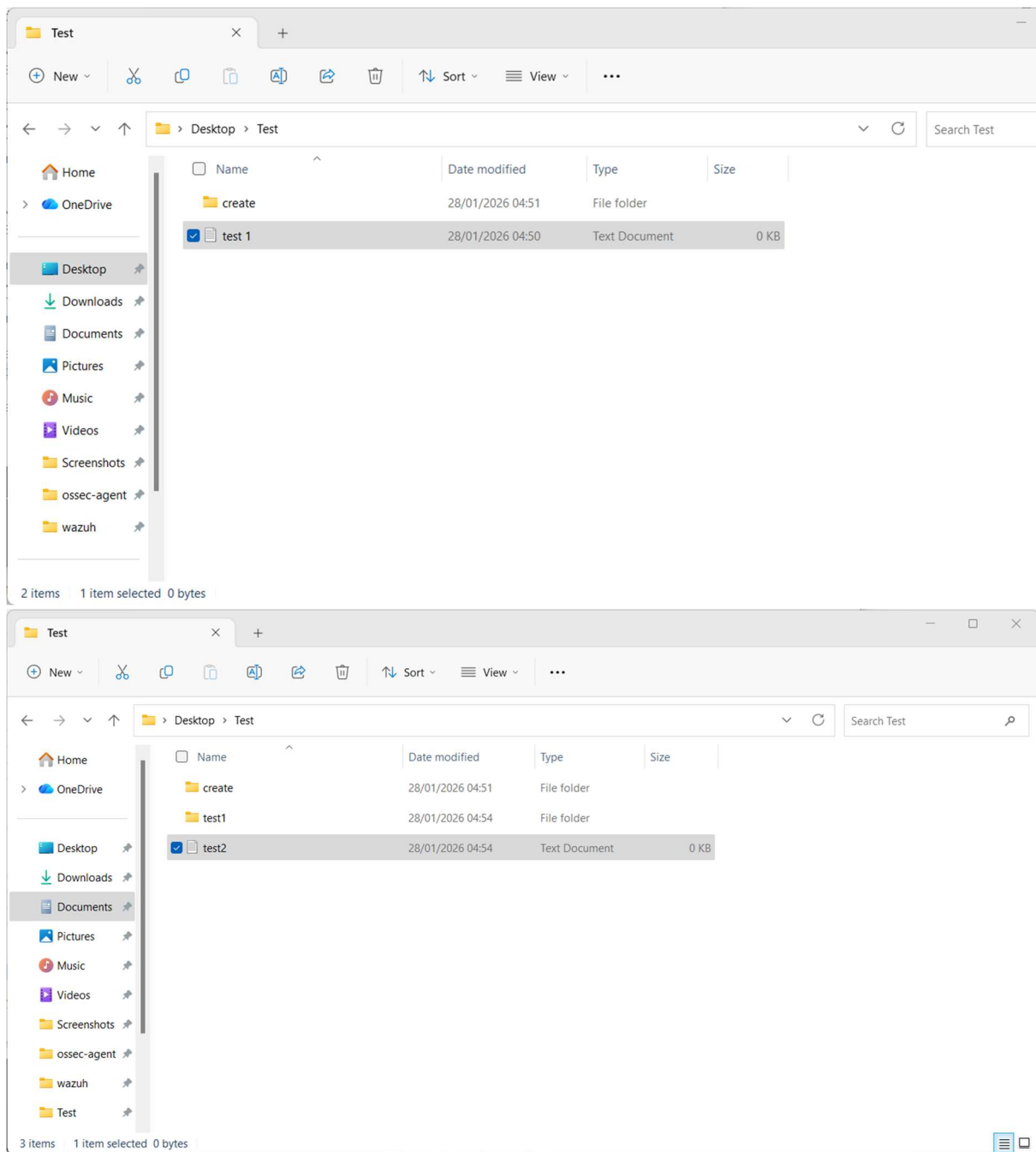


Figure 6 – Windows Explorer showing contents of Test folder before/after changes

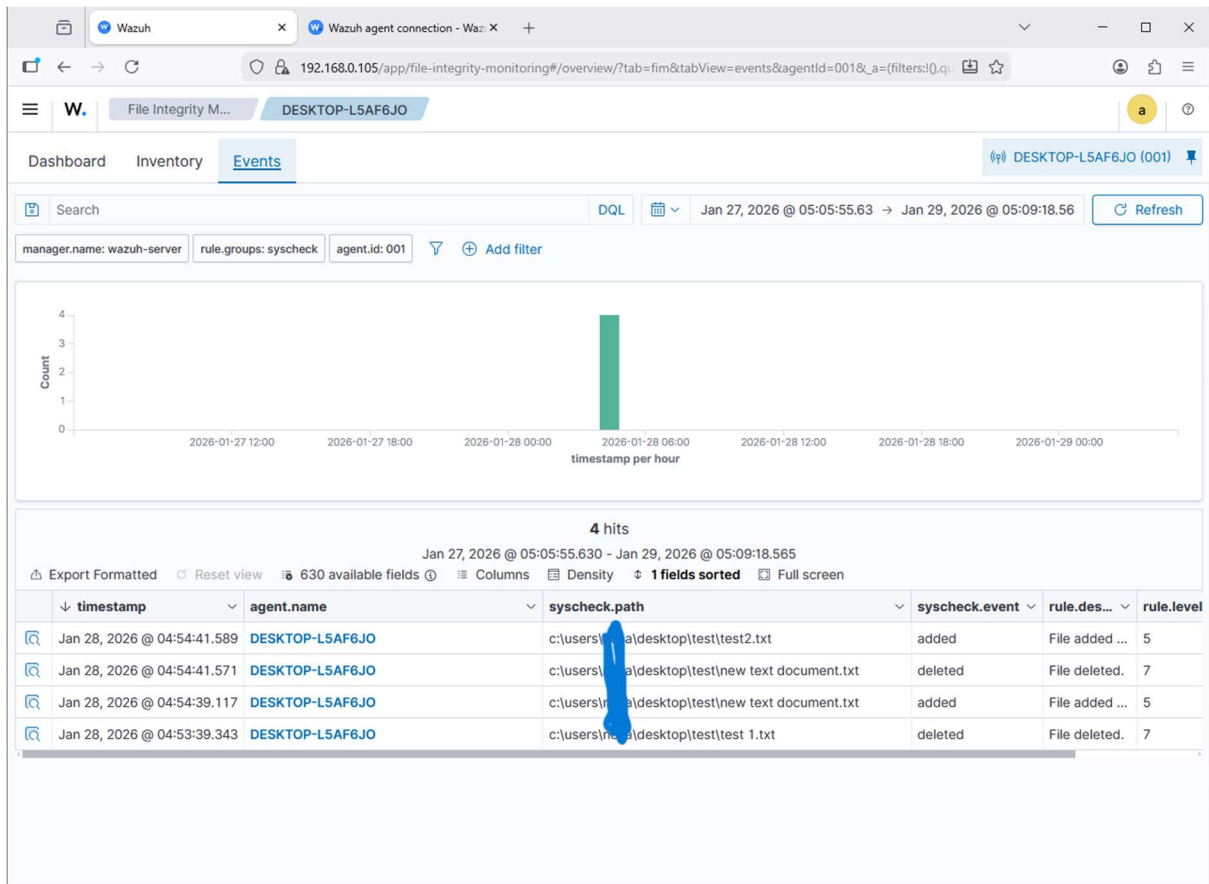


Figure 7 – Wazuh FIM Events table showing added and deleted files

The screenshot shows the Wazuh File Integrity Monitoring (FIM) Inventory table. The table displays a list of files and their properties. The data is as follows:

| File                                                      | Last modified               | User             | User ID          | Size   |
|-----------------------------------------------------------|-----------------------------|------------------|------------------|--------|
| c:\users\NOSA\desktop\test\test2.txt                      | Jan 28, 2026 @ 04:54:41.000 | NOSA             | S-1-5-21-2436... | 0      |
| c:\windows\regedit.exe                                    | May 7, 2022 @ 06:20:12.000  | TrustedInstaller | S-1-5-80-9560... | 552960 |
| c:\windows\system.ini                                     | Apr 12, 2018 @ 00:36:38.000 | SYSTEM           | S-1-5-18         | 219    |
| c:\windows\system32\drivers\etc\hosts                     | Apr 12, 2018 @ 00:36:39.000 | SYSTEM           | S-1-5-18         | 824    |
| c:\windows\system32\drivers\etc\lmhosts.sam               | May 21, 2023 @ 15:45:57.000 | SYSTEM           | S-1-5-18         | 3683   |
| c:\windows\system32\drivers\etc\networks                  | Apr 12, 2018 @ 00:36:39.000 | SYSTEM           | S-1-5-18         | 407    |
| c:\windows\system32\drivers\etc\protocol                  | Apr 12, 2018 @ 00:36:39.000 | SYSTEM           | S-1-5-18         | 1358   |
| c:\windows\system32\drivers\etc\services                  | Apr 12, 2018 @ 00:36:39.000 | SYSTEM           | S-1-5-18         | 17635  |
| c:\windows\system32\wbem\wmic.exe                         | May 10, 2023 @ 11:10:41.000 | TrustedInstaller | S-1-5-80-9560... | 520192 |
| c:\windows\system32\windowspowershell\v1.0\powershell.exe | May 10, 2023 @ 11:10:06.000 | TrustedInstaller | S-1-5-80-9560... | 491520 |
| c:\windows\system32\winrm.vbs                             | May 7, 2022 @ 06:19:27.000  | TrustedInstaller | S-1-5-80-9560... | 204074 |
| c:\windows\win.ini                                        | Apr 12, 2018 @ 00:36:38.000 | SYSTEM           | S-1-5-18         | 92     |

Figure 8 – Wazuh FIM Inventory listing monitored files (12 total)

## Step 6 – Simulate Privilege Escalation & Detect High-Severity Alert

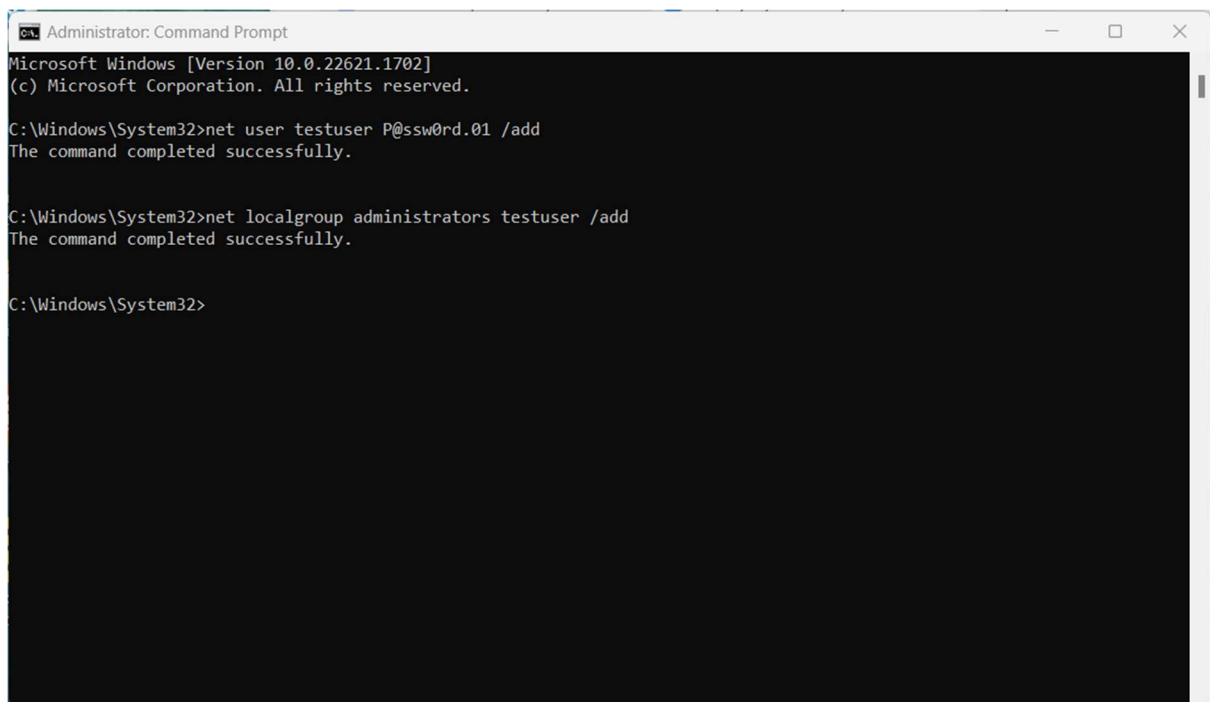
1. Opened elevated Command Prompt and ran:

```
net user testuser P@ssw0rd01 /add
```

```
net localgroup administrators testuser /add
```

2. Observed critical alert in dashboard → **Threat Hunting** → **Events**:

- **Rule 60154** – “Administrators Group Changed” → **Level 12** (very high severity)
- Related rules: 60170 (Users Group Changed), 60111/60110 (user account changed/created), 60109



```
Administrator: Command Prompt
Microsoft Windows [Version 10.0.22621.1702]
(c) Microsoft Corporation. All rights reserved.

C:\Windows\System32>net user testuser P@ssw0rd.01 /add
The command completed successfully.

C:\Windows\System32>net localgroup administrators testuser /add
The command completed successfully.

C:\Windows\System32>
```

Figure 9 – Command Prompt showing successful creation and addition of testuser to Administrators group

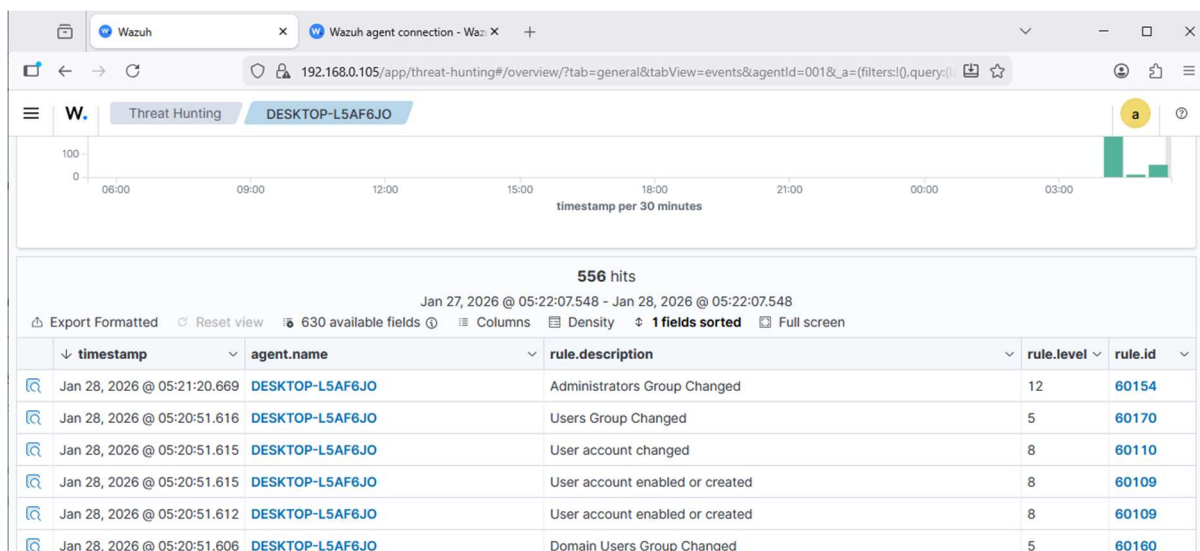


Figure 10 – Threat Hunting Events showing Rule 60154 Level 12 “Administrators Group Changed”

## Step 6A Add MITRE ATT&CK mapping

### MITRE ATT&CK Mapping

Several alerts generated during this lab map directly to MITRE ATT&CK techniques:

| Detected Behaviour                    | Wazuh Rule     | MITRE Technique ID & Name               | Tactic      | Notes                                    |
|---------------------------------------|----------------|-----------------------------------------|-------------|------------------------------------------|
| Administrators group changed          | 60154 (Lvl 12) | T1098 – Account Manipulation            | Persistence | High severity; immediate escalation risk |
| New local user created                | 60109 / 60111  | T1136 – Create Account                  | Persistence | Often combined with group addition       |
| File added/modified/deleted in folder | syscheck       | T1074.001 – Local Data Staging          | Collection  | Possible staging before exfiltration     |
| (If startup folder used)              | syscheck       | T1547.001 – Registry Run Keys / Startup | Persistence | Common persistence vector                |

## Step 7 – Review Authentication & Overall Alert Activity

1. Performed multiple logons (console and RDP attempts).
2. Reviewed dashboard → **Threat Hunting** overview:
  - Hundreds of events (sca, windows, ossec, windows\_security dominant)
  - Successful remote logons (Rule 92657 – level 6)

- Workstation logon success (Rule 60118 – level 3)

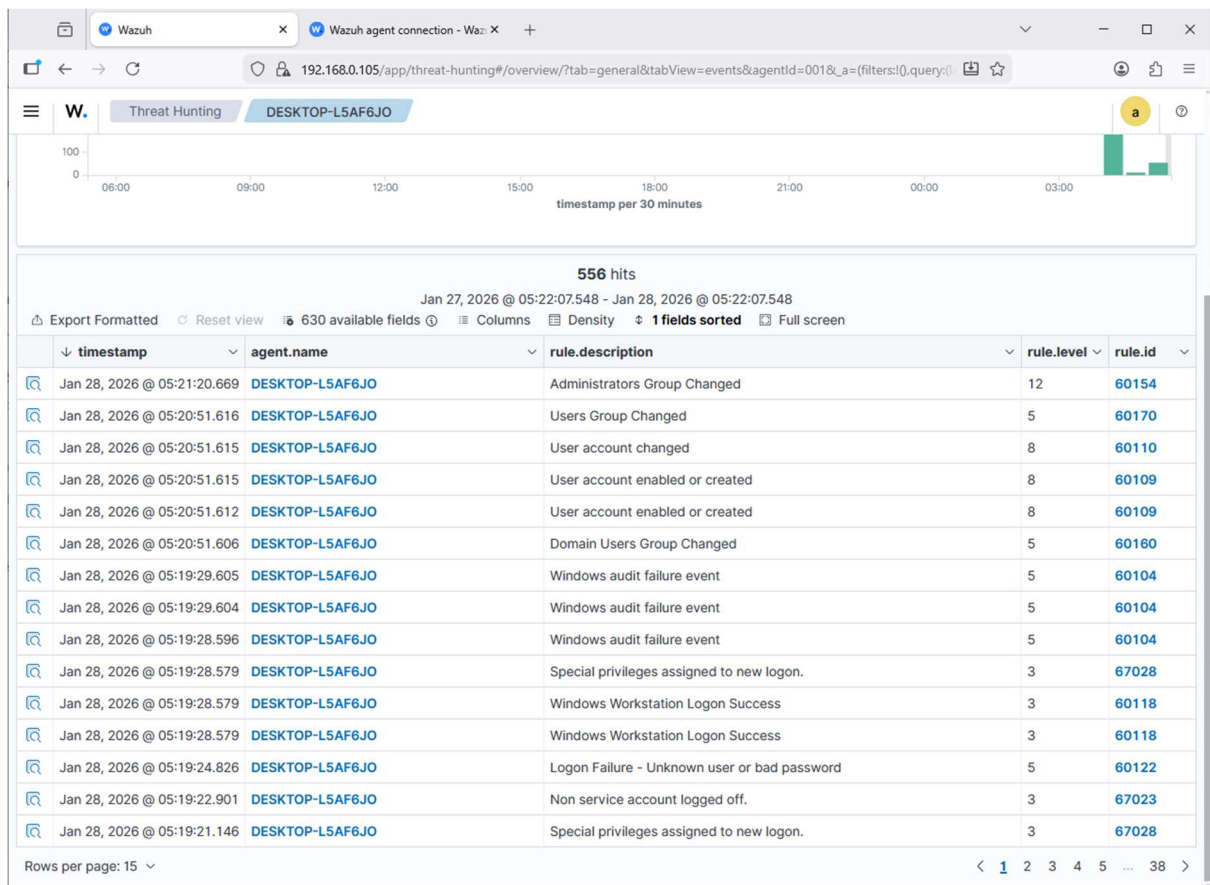


Figure 11 – Threat Hunting overview showing total alerts, top groups, and authentication success/failure counts

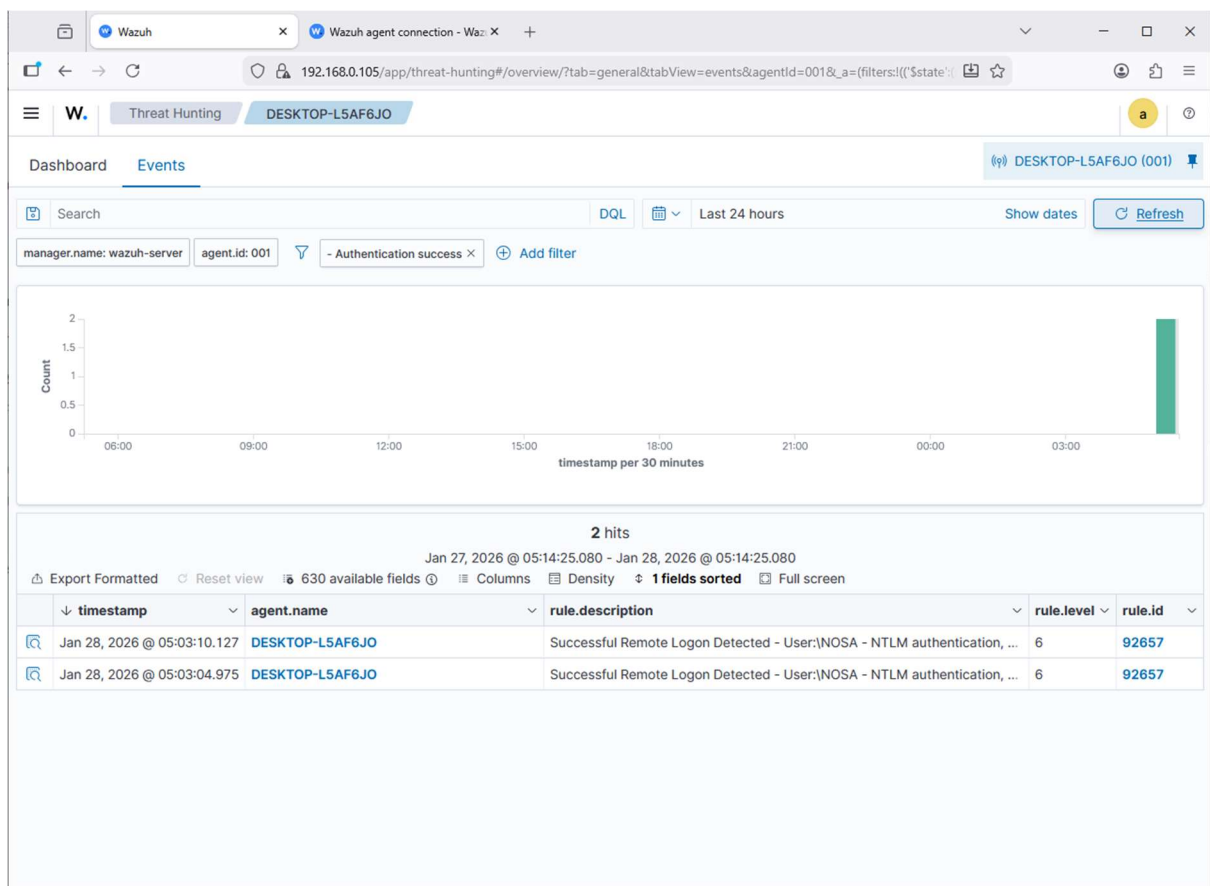


Figure 12 – Events filtered for “Authentication success” showing Rule 92657 remote logon detections

## Lessons Learned

- Changes to group membership (especially Administrators) trigger very high severity alerts and should always be treated as potential incidents.
- Realtime FIM is extremely useful for detecting unauthorized file activity, but path selection must be careful to avoid alert fatigue.
- Sysmon + Wazuh combination provides significantly better visibility than Windows logs alone.
- Documenting every step (commands, screenshots, timestamps) makes troubleshooting and reporting much easier.



## Next Steps & Future Work

- Write custom detection rules to reduce noise and catch specific behaviours
- Run Atomic Red Team simulations (e.g. T1059.001 PowerShell, T1105 tool transfer)
- Deploy a Linux agent and monitor Auditd / ssh logs
- Enable Active Response (auto-block IPs after brute-force)
- Explore cloud log sources (Microsoft Sentinel free tier)